ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 4

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 2 style (design-related, 80 marks), built around a 3-axis robot arm. Section A: objective questions (20) · Section B: power-supply design (linear and buck regulators, battery life, decoupling), H-bridge motor driver, encoders, level shifting and current sensing, and gripper logic with PLC ladder rungs and counters (60).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using the correct symbols. A ruler and a non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 An L298N H-bridge board is used to …

A control the speed and direction of DC motors.
B regulate the supply voltage.
C measure temperature.
D store data.

1.1.2 A 7805 regulator needs about 2 V more at its input than at its output, so its input must be at least …

A 5 V. B 7 V. C 12 V. D 3,3 V.

1.1.3 A buck converter …

A steps a DC voltage down efficiently.
B steps an AC voltage up.
C rectifies AC.
D amplifies audio.

1.1.4 The heat produced in a linear regulator is …

A $V_{out} \times I$.
B $V_{in} \div I$.
C $(V_{in} - V_{out}) \times I$.
D I only.

1.1.5 A logic-level shifter is needed when …

A 5 V and 3,3 V logic are connected.
B two batteries are joined.
C an LED is dimmed.
D current is measured.

1.1.6 Encoder pulses are best counted by using …

A delay(). B a fuse. C analogRead() only. D interrupts.

1.1.7 A Hall-effect current sensor often used in robots is the …

A LM35. B ACS712. C HC-SR04. D 555.

1.1.8 A PLC up-counter (CTU) counts …

A temperature. B voltage.
C time only. D the rising edges of an input.

1.1.9 A decoupling capacitor next to an IC …

A increases the voltage. B reduces noise on the supply.
C stores the program. D measures current.

1.1.10 A flyback diode across a motor protects the driver against …

A a low battery. B high temperature.
C back-emf voltage spikes. D bright light.

(10)

1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.2.1–1.2.10).

	COLUMN A		COLUMN B
1.2.1	Switching regulator that lowers a DC voltage	A	interrupt
1.2.2	Linear regulator with a fixed 5 V output	B	PWM
1.2.3	Four switches that can reverse a DC motor	C	buck converter
1.2.4	Signal whose on-time sets the average power	D	decoupling capacitor
1.2.5	Gives pulses as a motor shaft turns	E	encoder
1.2.6	Safely joins 5 V signals to 3,3 V inputs	F	overcurrent protection
1.2.7	Small capacitor at the supply pins of an IC	G	7805
1.2.8	Program event that runs immediately when an input changes	H	level shifter
1.2.9	PLC instruction that counts up to a preset value	I	CTU
1.2.10	Switches off a motor when its current is too high	J	H-bridge

(10)

TOTAL SECTION A: [20]

SECTION B

CASE STUDY. A college builds a 3-axis teaching robot arm. Each joint has a 12 V DC gear motor with an encoder, driven by H-bridges. The controller is an ESP32 (3,3 V logic). The arm has a 5 V object sensor and a gripper, and runs from a 12 V, 5 Ah battery.

QUESTION 2: POWER-SUPPLY DESIGN

2.1 Draw a neat block diagram of the power distribution: the 12 V battery and fuse feeding the motor rail, a 5 V rail for the sensors and a 3,3 V rail for the ESP32. (5)

2.2 A 7805 linear regulator supplies 0,8 A at 5 V from the 12 V battery.

2.2.1 Calculate the power wasted in the regulator. (2)

2.2.2 Calculate the efficiency of the regulator. (2)

2.2.3 Explain why a buck converter is a better choice. (2)

2.3 A buck converter with an efficiency of 90% supplies the same 5 V, 0,8 A load. Calculate the input power and the current drawn from the 12 V battery. (3)

2.4 The three motors draw an average of 0,6 A each and the electronics draw 0,4 A from the 12 V battery. Calculate how long the 5 Ah battery can run the arm. (3)

2.5 Explain why decoupling capacitors are placed close to the ESP32 and the sensor ICs. (3)

[20]

QUESTION 3: MOTOR DRIVER AND SENSOR INTERFACE

3.1 Explain how an H-bridge makes a DC motor turn forwards and backwards, and why the two switches on the same side must never be on together. (5)

3.2 Calculate the average voltage across a motor when the 12 V H-bridge runs at a duty cycle of 75%. (2)

3.3 Each motor's encoder gives 360 pulses per motor revolution and the gearbox ratio is 30 : 1. Calculate:

3.3.1 The number of pulses for one revolution of the joint (2)

3.3.2 The joint angle for each pulse (2)

3.4 The 5 V object sensor must be connected to a 3,3 V ESP32 input with a voltage divider made of 1,8 kΩ and 3,3 kΩ. Calculate the input voltage at the ESP32 when the sensor output is 5 V. (3)

3.5 An ACS712 current sensor gives 2,5 V at 0 A and 185 mV per ampere. It reads 2,87 V. Calculate the motor current. (3)

3.6 Explain why the encoders should be read with interrupts rather than by checking them in the main loop. (3)

[20]

QUESTION 4: DIGITAL AND PLC CONTROL DESIGN

4.1 The gripper G must close only when an object is detected (O = 1) AND the arm is at the pick position (P = 1) AND there is NO fault (F = 0).

4.1.1 Write the Boolean expression for G. (2)

4.1.2 Draw the logic circuit for G. (4)

4.2 In the PLC version, once the gripper has closed it must stay closed (seal-in) until the release button R is pressed or a fault occurs. Draw the ladder rung for G. (5)

4.3 The PLC must count the parts placed in a box with an up-counter (CTU) preset to 10. When the count is reached, a "Box full" lamp must light, and the RESET button must reset the counter. Describe or draw the rungs needed. (5)

4.4 Explain the purpose of a watchdog timer in the arm's controller. (2)

4.5 Explain why the emergency stop must be hard-wired and not only handled by the program. (2)

[20]

TOTAL SECTION B: [60]

TOTAL: 80

END OF QUESTION PAPER